Assignment: Analysis of Functional Connectivity from Resting-state fMRI [Without using fmriprep Pipeline]

🎯 Objective

To preprocess and analyze resting-state fMRI (rs-fMRI) data acquired from the **ADNI repository** and perform **functional connectivity analysis** using classical neuroimaging software tools such as:

- **FSL**
- **ANTs**
- **FreeSurfer**
- **NiPy / Nilearn**
- **NiBabel**
- **AFNI tools (optional)**

Background

Resting-state fMRI is used to study **functional connectivity** (synchronous activity between brain regions when no task is performed). This assignment requires applying a complete preprocessing pipeline and FC analysis *without automatic pipelines like fMRIPrep*.

📂 Dataset

Download the following from **ADNI (Alzheimer's Disease Neuroimaging Initiative)**:

- Resting-state fMRI (minimum 5 subjects: CN vs MCI or AD)
- Corresponding **T1-weighted structural MRI**

Tools Required

Category	Tools
Operating System	Linux / WSL / macOS
Neuroimaging Software	**FSL, ANTs, FreeSurfer**
Python Libraries	Nibabel, Nilearn, Nipype, NumPy, Pandas, Matplotlib
Optional	AFNI utilities, MRIcron, ITK-SNAP

✅ Preprocessing Pipeline (Manual, No fMRIPrep)

A) Structural MRI Preprocessing

Step	Tool
Convert DICOM → NIfTI	`dcm2niix`

Step	Tool
Bias-field correction	ANTs `N4BiasFieldCorrection`
Brain extraction	**FreeSurfer `recon-all`** OR **FSL BET**
Segmentation (WM/GM/CSF)	FreeSurfer / FSL FAST
Surface reconstruction	FreeSurfer

B) fMRI Preprocessing

Step	Command/Tool
Remove first 5–10 volumes	FSL `fslroi`
Motion correction	FSL `M CFLIRT`
Slice-time correction	FSL `SLICETIMER`
Coregister fMRI → T1	FSL `FLIRT` or ANTs `antsRegistration`
Normalize to MNI template	ANTs `antsApplyTransforms`
Spatial smoothing (6 mm FWHM)	FSL `SUSAN`
Temporal filtering (0.01–0.1 Hz)	FSL `fslmaths` bandpass
Denoising (WM/CSF confounds)	FreeSurfer masks + regression (Nilearn/Nipype)

C) Functional Connectivity Analysis

Perform **Seed-Based or ICA FC analysis** using FSL:

Task	Tool
Seed-based connectivity	FSL `FEAT` + custom ROI
Group ICA	**MELODIC**
Network map visualization	FSLeyes, Nilearn
Connectivity matrix	Nilearn `ConnectivityMeasure`

Suggested ROI seeds:

- PCC (Posterior Cingulate Cortex) — Default Mode Network
- Hippocampus — Memory network (AD-relevant)

📊 Expected Outputs

✔ Preprocessed fMRI images
✔ Motion plots
✔ Seed-based connectivity maps
✔ Group ICA maps (MELODIC)
✔ CSV Correlation matrix between brain regions
✔ Report describing network findings (DMN differences in AD vs CN)

✍ Submission Requirements

Deliver the following:

1. Report (PDF)

- Dataset description
- Full preprocessing steps & screenshots
- FC analysis results
- Brain maps & interpretation

2. Scripts

- Shell scripts for FSL & ANTs commands
- Python scripts (Nilearn/Nibabel/Nipype)

3. Outputs Folder

- Preprocessed brain images
- Network maps
- Correlation matrices

🎯 Grading Rubric

Criteria	Marks
Preprocessing pipeline	30
Correct tool usage (no fMRIPrep)	20
FC results and interpretation	25
Code + reproducibility	15
Presentation/report	10

✅ Outcome

After completing this assignment you will:
✔ Understand complete rs-fMRI preprocessing workflow
✔ Perform connectivity analysis (seed-based & ICA)
✔ Work with real ADNI neuroimaging data
✔ Use classical neuroimaging tools used in research labs
